VaxImpact Simulator – Student Guide

How to use the simulator and what you will do in the worksheet

1. YOUR GOAL

You will use a computer simulation to investigate how changing vaccination rates can affect disease spread. You will make a prediction, run repeated trials, collect evidence, graph your results, and use the evidence to explain a pattern.

Remember

VaxImpact is a simplified model of disease transmission. It helps you investigate how certain variables—including vaccination rate, population density, vaccine effectiveness, transmission rate, mortality rate, and infectious period—influence disease transmission.

2. LEARN THE SIMULATOR BEFORE YOU EXPERIMENT

Feature and Location	What it does and what you should do
Main simulation panel - center	Shows the modeled spread of infection. Watch how the community changes as the simulation runs.
Controls - beside the simulation	You can change vaccination rate, transmission rate, vaccine effectiveness, recovery time, mortality rate, and population density.
Play	Starts the model. Let the run finish before you record final results.
Speed controls - top of simulator after Play	Use these to make the simulation run faster or slower. They change speed, not your experiment settings.
Epidemic Timeline - scroll down after Play	Find peak infections, total infected, % vaccinated sick, and % unvaccinated sick. These are the values you record in your worksheet.
Learn More - upper-right area	Use this when you need a clearer explanation of a variable.

3. HOW THE WORKSHEET FLOWS

Part	What you will do
Part I - Explore & Question	Explore the simulator by changing different variables and observing how the outbreak changes. Then write a testable question and identify the independent variable, dependent variable(s), and control variables.
Part II – Outbreak Challenge	Use the simulator to create the **largest and smallest outbreak** you can while following the challenge rules. Record your settings and results, then identify which variables seemed most important.
Part III – Controlled Investigation	Make a prediction about how vaccination rate will affect the outbreak. Test **65%, 80%, and 95% vaccination**, running three trials at each level while keeping all other settings constant. Record and average your results.
Part IV – Analyze, Graph, and Explain	Organize your average results and graph total and peak infections across vaccination rates. Decide whether your prediction was supported, then make a **claim** and support it with **evidence and reasoning** from your results.
Part V – Cause and Effect Reasoning	Use the simulator and your evidence to explain how **vaccine effectiveness, population density, and recovery time** can affect disease spread. Think about community effects of vaccination and identify a limitation of the model.
Part VI – Connect the model to the Real World	Read a credible source about a real disease outbreak or vaccination. Identify important facts, connect them to patterns you observed in the simulator, and explain why real outbreaks can be harder to predict than a computer model.

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4. HOW TO RUN A TRIAL

Step	What to do
1. Set the controls	For Part II, use the baseline settings listed in the worksheet.
2. Set vaccination rate	Choose the condition you are testing: 65%, 80%, or 95%.
3. Press Play	Start the simulation and let the model run. You may also change the speed to help this portion finish sooner.
4. Scroll to the Epidemic Timeline	Read the requested outcome values after the run finishes.
5. Record the results	Enter the values in the correct trial row. Then repeat until you have three trials for that vaccination rate.
6. Look for patterns	Compare the three trials and calculate/use the average required by the worksheet.

5. Definitions of Scientific Terms and Principles

Scientific Terms & Principles	Definition
Independent Variable	The factor that is intentionally changed or manipulated in an experiment.
Dependent Variable	The outcome that is measured to determine how it responds to changes in the independent variable
Control Variables	Factors that are kept the same between conditions so that differences in the results can be more confidently linked to the independent variable.
Trial	One complete run of the simulation or experiment using a specific set of conditions.
Repeated Trials	Repeating the same condition several times to determine whether a pattern is consistent rather than relying on a single result.
Data	Measurements or observations collected during an investigation.
Average/Mean	A value found by adding several results together and dividing by the number of results.
Claim	A conclusion or statement that answers the scientific question being investigated.
Evidence	Relevant data or observations used to support or evaluate a claim.
Reasoning	The explanation of why the evidence supports the claim using scientific ideas or principles.
Vaccination Rate	The percentage of the population who are vaccinated.
Vaccine Effectiveness	A measure of how much vaccination reduces the risk of a particular health outcome compared with not being vaccinated. CDC notes that vaccine effectiveness can refer to protection against outcomes such as infection, illness, hospitalization, or death. Learn more here: https://www.cdc.gov/flu-vaccines-work/php/effectivenessqa/terminology.html
Transmission Rate	A measure in the simulation that represents how readily an infection spreads from infected individuals to susceptible individuals.
Infectious Period/	How long an infected person remains infected and able to spread the disease.

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Recovery Time	
Population Density	People per unit of area or how closely people are located to one another in the simulated population. This influences the rates of and opportunities for contact.
% Vaccinated Sick	The percentage of Vaccinated Individuals who became infected at any point during the simulation.
% Unvaccinated Sick	The percentage of Unvaccinated Individuals who became infected at any point during the simulation.
Peak Infections	The greatest number of people infected at the same time during the simulation/outbreak.
Total Infected	The total number of different people who became infected at any point during the simulation/outbreak.
Strong Evidence	Evidence based on specific data, comparisons, and consistent patterns across repeated trials rather than a single unusual result.
Outbreak	An increase in cases of a disease within a population or area.
Indirect Protection	Protection that can occur when vaccination reduces disease spread, lowering the chance that other people will be exposed. CDC also uses terms such as "community protection," "herd protection," and "indirect effects." Read more here: https://www.cdc.gov/flu-vaccines-work/benefits/index.html
Model	A simplified representation of a real system that can be used to investigate patterns, relationships, and possible outcomes.
Model Limitation	A feature of the real world that a model simplifies, leaves out, or cannot represent perfectly.
Model Thinking	Models can help you understand a complex system while still leaving out important real-world factors. When you make a claim, think about what the simulator includes and what it does not.

Before you submit the assignment, take a brief sanity check and make sure you did the following.
□ I identified my variables. □ I made a directional prediction. □ I completed repeated trials. □ I used the Epidemic Timeline correctly. □ My graph has a title, labeled axes, and units. □ I described the trend. □ I used evidence in my explanation. □ I considered a model limitation.